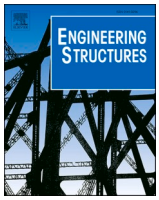
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A digital twin framework for damage detection and localization of self-sensed cured-in-place underground pipelines

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ABSTRACT

As the underground pipelines deteriorate with age, their structural vulnerability increases. Cured-in-place pipe (CIPP) lining is an emerging trenchless technology used to rehabilitate deteriorated pipes. However, the buckling behavior of CIPP liner-rehabilitated pipes is insufficiently investigated. This study presents an integrated framework for real-time structural health monitoring and damage detection for underground pipelines by leveraging the knowledge of distributed fiber optic sensors (DFOS), finite element analysis (FEA), deep learning (DL) algorithms, and digital twin. The study experimentally and computationally investigates the structural behavior of a single steel substrate and a CIPP-rehabilitated steel substrate under eccentric compression conditions. The application of DFOS enables the self-sensing function of the pipelines with high-resolution, continuous strain measurement. The finite element model and DL-based strain field correlation are employed to complement the function of DFOS from strain field monitoring to structural deformation assessment, facilitating the development of virtual replicates of self-sensed pipes. The study further explores the location and severity of structural damage through numerical and theoretical strain analysis, enabling the detection of minor but critical defects. The developed digital twin framework enables real-time structural visualization and enhances structural damage assessment. By accurately identifying the early-stage plastic region of deformation and quantifying crack severity, the framework supports predictive maintenance and improves pipeline reliability. This proactive approach improves the safety of underground pipelines, extends their service life, and reduces maintenance costs, demonstrating its practical potential for long-term pipeline infrastructure management.

1. Introduction

Underground pipeline networks form the backbone of modern urban infrastructure, enabling the efficient transportation of water, gas, sewage, and various industrial fluids [1]. The underground pipeline systems are essential to economic growth, public health, and environmental protection. However, with the progression of time, underground pipelines become increasingly susceptible to deterioration due to material degradation, external loads, corrosion, and soil-structure interactions [2]. The failure of underground pipelines can result in severe economic losses, environmental contamination, service disruptions, and even loss of life. Consequently, ensuring the long-term structural integrity of underground pipelines has become a critical challenge in civil and infrastructure applications. Conventional methods, such as replacing deteriorated pipelines, often result in substantial inconvenience to local residents and industrial facilities due to service interruption. In contrast to conventional methods, various trenchless technologies have been developed to restore the function of pipelines while minimizing excavation and associated disruptions [3]. Among them, the cured-in-place pipe (CIPP) lining method has gained widespread recognition for its adaptability to different pipe materials and sizes, cost-effectiveness, and capacity to restore the structural integrity of deteriorated pipes [4].

The CIPP lining method involves inserting a resin-based liner material into an underground pipe. After being cured in place appropriately, the liner effectively forms a new structural layer within the host pipe [5]. Enabled by the enhanced structural integrity provided by the CIPP liner, this method enhances pipeline durability, improves hydraulic efficiency, and extends the service life of the liner-rehabilitated pipe system. Despite these advantages, the liner-rehabilitated pipe system can

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still experience structural failures. Common damage mechanisms include cracking, delamination, and deformation due to internal or external interferences, with buckling being one of the most critical failure modes [6–8]. Buckling in underground pipelines, whether original or rehabilitated with a liner, can be caused by soil loads, groundwater pressure, traffic-induced stresses, or vacuum-induced forces [9–12]. Since such failures often occur suddenly and can lead to critical structural instability, early detection and mitigation are essential for ensuring the pipeline's reliability. Prior studies have evaluated the structural performance of CIPP liners through experimental testing and analytical or numerical modeling, including investigations of ovalization, local buckling instability, liner–host pipe interaction, and critical capacity [13–16]. These studies have improved the understanding of the progression of buckling from initial imperfection to post-buckling deformation in liner-rehabilitated pipes. However, existing investigations generally rely on visual inspection or discrete sensors, which are unable to capture the distributed strain evolution that governs buckling initiation. Therefore, real-time tracking of buckling behavior and deterioration in liner-rehabilitated systems remains an open challenge, due to the inaccessibility of pipelines and the complexity of their operating environment.

Structural health monitoring (SHM) is an approach to assess the condition of critical infrastructure, providing valuable insights that can help detect early signs of damage and prevent catastrophic failures [17–19]. However, traditional SHM techniques face significant challenges when applied to underground pipelines. Conventional point-based sensors, such as strain gauges or fiber Bragg grating (FBG) sensors, are difficult to deploy in underground environments and offer limited spatial coverage. Therefore, these sensors are not suitable for monitoring underground pipelines, where the locations of potential failures are unpredictable. To address these challenges, distributed fiber optic sensing (DFOS) has emerged as a promising solution to monitor the structural health condition of underground pipelines in real-time [20]. Unlike conventional sensors, DFOS enables spatially continuous strain measurements along the entire length of the optic fiber. This continuous coverage allows for real-time strain monitoring over a large area. Anomalous strain signals are indicative of structural irregularities, enabling both damage identification and localization. As a result, DFOS facilitates the timely detection of localized damage, enables the assessment of overall strain distribution, and monitors structural performance with higher resolution and accuracy. Additionally, DFOS offers several advantages, including its ability to work in harsh environments, its high sensitivity to strain variations, and its long-term stability, making it an ideal choice for underground applications. Recent studies have employed DFOS as a strain monitoring tool for various underground applications, including fractured steel pipelines subjected to operational pressure [21], polyvinyl chloride liners under grouting pressure [22], and corrugated steel culverts under traffic loading [23]. These previous studies demonstrate the capability of DFOS to provide continuous strain monitoring in underground structures, establishing a foundation for its effective integration into real-time SHM applications. The effectiveness of DFOS relies on its underlying principle of analyzing backscattered light along the optical fiber. In an optical distributed sensor interrogator, a laser source with a continuously varying frequency is injected into the fiber, and the backscattered Rayleigh signal is analyzed. Since the Rayleigh scattering pattern is unique to each segment of the fiber, any mechanical strain changes the local optical path, resulting in a shift in the backscattered signal. By comparing these shifts with the reference state, highly precise distributed strain measurements can be obtained at sub-millimeter-scale spatial resolution (SR). This technique makes Rayleigh-based DFOS particularly suitable for detecting fine-scale strain variations and tracking minor defect initiation and propagation. As a result, it is highly promising for SHM applications of underground pipes, where direct visual detection is often infeasible.

DFOS provides a powerful tool for monitoring the structural health condition of underground pipelines. However, it is unrealistic to have DFOS fully cover the entire pipe due to the high cost of fiber optic sensors. Additionally, raw sensing data alone may not fully capture the complexities of pipeline behavior under real-world conditions. Therefore, in practical deployment, DFOS is installed on instrumented segments that terminate at accessible interrogation points. Since pipeline distress typically occurs locally, the sensing data from these instrumented segments are used to update finite element models of the broader system. By combining localized measurements with numerical predictions, the integration of DFOS and finite element analysis (FEA) provides a comprehensive assessment of the structural performance of the underground pipeline system. FEA-based numerical modeling enables the simulation of various loading conditions, damage mechanisms, and failure modes, allowing for the prediction of how a pipeline will behave under different scenarios [24,25]. This approach shows a promising approach to conduct virtual testing on pipelines and generate a dynamic, real-time digital replica through digital twin technology. In the context of infrastructure, the digital twin framework can continuously update and provide a virtual representation that offers direct insight into the structural condition [26]. Wang et al. [27] applied polynomial fitting to process experimental and numerical strain data and reconstruct the structural deformation of a pipe-liner system in a three-dimensional digital twin model. Wang et al. [28] developed a digital twin system for pipeline fatigue crack monitoring through the internet of things, advanced analytics, cloud computing, and virtual reality. However, there is a lack of efficient and timely digital twin modeling methods for SHM in pipeline systems, as current practices require substantial computational time for data post-processing. To address this issue, the incorporation of machine learning (ML) algorithms represents a further step toward real-time data processing between the physical twin and the digital twin [29]. Limited studies have advanced digital twin-based SHM using ML approaches in infrastructure domains, most of which focus on bridge components [30–32], monitoring deflection, moment, and crack propagation. Similar advancements have not yet been fully realized for underground pipeline infrastructure applications. Building on these established practices, extending the use of ML toward continuous condition monitoring of pipeline structures, enabling more effective and preventive SHM strategies, is necessary.

In order to enhance the effectiveness of SHM for underground pipeline infrastructure, it is important to enable preventive damage identification and localization. Hoult et al. [33] created artificial damages in steel plate samples and detected subsequent deterioration using DFOS by measuring strain variations. Lan et al. [34] installed DFOS on a perforated steel pipeline to measure the longitudinal strain distribution under ground movement. They found that while the global behavior of the perforated pipeline remained unchanged, the strain variation around the perforation was significant, highlighting the capability of DFOS for damage identification in underground pipeline applications. However, these studies were based on predefined, visually identifiable damages. To bridge this gap, DFOS needs to be utilized to continuously monitor the full structural damage evolution, from initially intact conditions through progressive deterioration leading to failure, enabling comprehensive SHM of damage propagation and facilitating early-stage damage assessment. One key component of structural damage originates from the plastic deformation of steel, which induces permanent deformation. Once the localized strain exceeds the yield strain, the affected region enters plasticity. This plastic region of deformation cannot restore its original shape after the external load is removed, resulting in permanent deformation. This irreversible deformation compromises the strength, bearing capacity, and transmission efficiency of the underground pipe [35]. Another key focus is the investigation of crack initiation at critical structural interfaces, such as the adhesive layer. The adhesive layer is the intermediate layer between the pipe and CIPP liner, used to bond the liner firmly to the host pipe [36]. This layer has been largely overlooked in previous studies despite its impact on the overall rehabilitation performance of the CIPP liner. Large structural

deformation, such as buckling, can result in localized defects in the adhesive layer, compromising the integrity of the liner-rehabilitated pipe system. Therefore, it is essential to identify the conditions and locations of such cracks. In this context, integrating DFOS with the CIPP liner is suggested as a potential solution to monitor the condition of the entire liner-rehabilitated pipe. While the installation of DFOS directly on the underground pipe remains challenging, embedding DFOS within the liner prior to installation offers a promising alternative. Since DFOS is not directly attached to the pipe surface, the strain monitoring results may be affected, making it critical to investigate the performance of DFOS-integrated liners in effectively monitoring pipe conditions.

The objective of this research is to develop a digital twin framework for real-time monitoring and damage identification in underground pipelines. This work seeks to obtain real-time strain fields from DFOS-embedded steel substrate samples, develop corresponding finite element models to characterize buckling behavior and structural responses, correlate DFOS-measured and FEA-predicted strain fields to update the virtual pipeline state, and identify structural damage such as plastic deformation and adhesive cracks through strain-based analysis. Section 2 introduces the methods of this study through five main steps for developing the digital twin framework. First, the testbed for samples subjected to eccentric compressive loading is introduced, and strain distributions on the samples are collected in real-time by DFOS. Second, one-dimensional DFOS strain data are converted into two-dimensional strain fields across the sensor deployment area. Third, finite element models are developed to simulate the experimental setup, and nonlinear analysis is performed to track structural buckling deformation and variations in strain distribution. Fourth, to connect the strain field with the structural response, a deep learning (DL)–based strain field correlation analysis is performed between DFOS measurements and FEA predictions, enabling the selection of the best-matching strain fields and the employment of corresponding FEA structural response results to generate the digital twin framework. Lastly, structural damage, including plastic regions of deformation and adhesive cracks, are identified and localized through numerical and theoretical strain analysis. The damage information is incorporated into the digital framework to enrich the information presentation and facilitate the virtual visualization of the holistic structural health condition. Section 3 presents the results and discussion from the developed digital twin framework for the smart liner system. The innovative approach presented in this study advances the field of underground pipeline infrastructure by enabling real-time SHM, structural performance prediction, and automated damage identification. Through continuous updates of the pipeline's virtual state, the digital twin framework supports proactive maintenance, reduces unplanned service disruptions, and enhances risk management for aging systems. The research contributes to the development of a self-sensed, cured-in-place pipeline system that enhances the safety, durability, and operational reliability of underground networks.

2. Digital-twin framework development for buckling

This section aims to establish a digital twin framework for damage detection and localization in self-sensed, cured-in-place underground pipelines. Fig. 1 shows the procedures to develop the framework for a virtual, dynamic, and real-time visual representation of the pipe's behavior. A testbed is developed to experimentally investigate the eccentric buckling behavior of the CIPP liner-pipe system. With the application of distributed fiber optic sensing technology, the pipe becomes "self-sensing", with real-time generation of high-resolution strain measurements. After the strain data are collected by the DFOS interrogator, they are converted into coordinate-based strain results in terms of the SR and corresponding geometric location. This conversion enables the generation of a two-dimensional experimental strain field. The experimental strain field is then systematically compared with the numerical strain field obtained from FEA to validate the finite element model. A DL-based method is applied to extract and correlate the features from both DFOS and FEA strain fields throughout the entire eccentric compression process. This correlation helps real-time identification of underlying structural mechanical behaviors from strain distributions and supports the visualization of structural deformation via the digital twin framework. To further enhance the versatility and diagnostic capacity of the digital twin framework and fully leverage the potential of DFOS, the damage regions on the samples are identified and localized. The developed digital twin framework enables the visualization of two types of commonly invisible damage: plastic deformation in the steel substrate and cracking in the adhesive layer. The visualization of plastic deformation provides insight into permanent, irreversible damage on the steel substrate, and the identification of adhesive cracks supports the timely diagnosis of the rehabilitation performance of the CIPP liner. The innovative integration of real-time sensing technology, finite element modeling, and deep learning algorithms develops this enhanced digital twin framework, enabling the capacity to generate a real-time, visual representation of structural and mechanical behaviors, as well as structural damage in the steel substrate.

The real-time capability of the proposed digital twin framework is enabled by a pre-trained inverse workflow. High-resolution DFOS continuously monitors spatial strain variations, which are instantaneously converted into two-dimensional strain fields based on the predefined sensor spacing and spatial resolution. A comprehensive nonlinear FEA database is pre-generated, and deep-learning algorithms rapidly correlate the incoming DFOS strain fields with the numerical database to reconstruct full-field deformation and identify damage regions without iterative simulations. Damage identification and localization are performed automatically, as spatial strain gradients and their temporal evolution indicate cracks and plastic regions of deformation without manual intervention. This framework facilitates real-time, comprehensive SHM and predictive maintenance of underground pipelines, providing a clear path to directly visual representation for assessing potential failure mechanisms before they propagate into critical issues.

2.1. Testbed for buckling behavior of the self-sensed cured-in-placed pipe liner-pipe system

In order to reflect and investigate the feasibility of the self-sensed cured-in-place underground pipelines in practice, three layers,

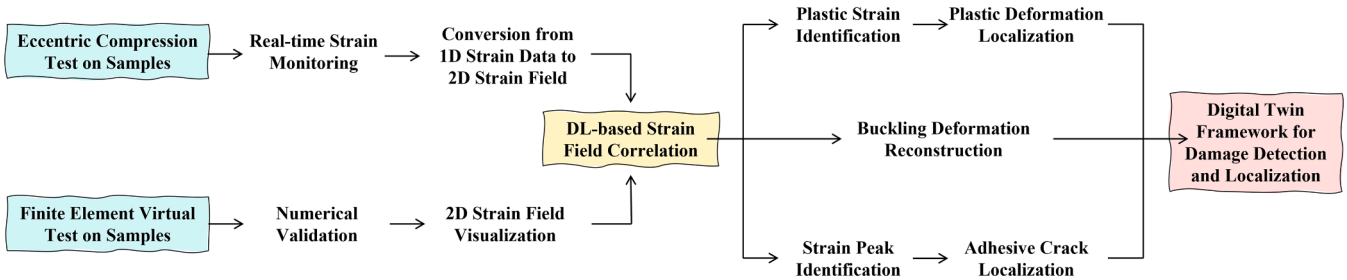


Fig. 1. Procedures to develop the digital twin framework for damage identification.

including the host pipe, adhesive layer, and CIPP liner, are selected in this study. The materials used for the development of test samples include steel substrate, epoxy resin adhesive, CIPP liner, and DFOS. A flat steel substrate is used to simulate the underground pipe. Epoxy resin adhesive is a thermosetting resin, and is commonly used to form an adhesive layer between the pipe and CIPP liner, establishing a robust bond to ensure the effective rehabilitation performance. The CIPP liner, made of fiber-reinforced polymer composites, typically consists of polyester or epoxy resin reinforced with glass or carbon fibers, and polyurethane/polyethylene coating. A commercial CIPP liner (Starline® 2000), a widely applied product for rehabilitating deteriorated underground pipes in the U.S., is used in this study [37].

To achieve the function of self-sensing, DFOS is installed on the samples for real-time, continuous strain monitoring. Distributed fiber optic sensing is a sensing technology that utilizes optical fiber to continuously measure strain along the entire fiber length. This allows for high SR and accurate strain monitoring of structural elements. The optical fiber, typically made of silica glass, serves as both the sensing element and the signal transmission medium [38]. The DFOS system used in this study operates based on Rayleigh scattering within the fiber. Rayleigh scattering occurs due to the microscopic density and compositional variation in the fiber material, causing a small fraction of the transmitted light to scatter in all directions [39]. When the backscattered signal interferes with the reference signal generated by the interrogator (Luna ODiSI 6100 in this study), it generates an interference pattern containing phase and amplitude information, which correlates with strain variations along the optical fiber. By applying a fast Fourier transform, the system extracts high-resolution, spatially continuous strain data. This processed data is used to generate a high-resolution strain profile along the fiber length, providing real-time, visually accessible graphical strain feedback on the control platform. The self-sensing capability enabled by DFOS significantly enhances the awareness of structural issues in underground pipe infrastructure.

As shown in Fig. 2(a), the DFOS is embedded on the surface of a 152.4 mm × 304.8 mm × 0.81 mm (W × L× H) steel substrate sample in a zig-zag pattern to maximize area coverage. This installation pattern enlarges the monitoring area and ensures comprehensive strain measurement of the substrate under buckling conditions. Since the direct installation of DFOS on in-service underground pipes is difficult to implement, one solution is to pre-embed the DFOS in the CIPP liner before its installation into the underground pipe, thereby enabling the self-sensing capacity. Additionally, the integrity of the adhesive layer directly determines the effectiveness of CIPP rehabilitation; any crack or defect is important to be detected in a timely manner. Therefore, a same-sized liner-rehabilitated steel substrate sample is developed, which has two DFOS embedded at the interface between the steel substrate and adhesive layer and the interface between the adhesive layer and CIPP liner, as shown in Fig. 2(b). To fabricate this sample, two DFOS are installed on the surfaces of the CIPP liner and steel substrate with thin layers of epoxy resin adhesive. The sensors are installed in the same area and in the same pattern on these two layers to ensure consistency. After curing at room temperature for 6 h, the DFOS-installed CIPP liner and steel substrate are bonded together firmly using a thick layer of the aforementioned epoxy resin adhesive. The prepared sample is then placed at room temperature for a 24-hour curing period before testing. The thickness of each layer from top to bottom is 1.94 mm, 1 mm, and 0.81 mm, respectively. To investigate the behavior of the underground pipe under complex loading conditions, an eccentric compressive load is applied to the test samples through an MTS Criterion® electromechanical test machine. The top grip clamps the upper corner of the sample over an area of 25.4 mm × 30 mm (W × L) and monotonically moves downwards at a constant rate of 0.003 mm/s. The bottom grip fixes the bottom corner with the same dimensions to ensure consistency. The total displacement is 7 mm at the first stage, simulating a significantly large buckling deformation, with the maximum out-of-plane deformation approximately 20 mm. Moreover, underground pipes can experience complex loading conditions that result in multiple plastic deformations. Therefore, a second eccentric compressive test with a 4 mm displacement is conducted on the single steel substrate to investigate its performance after initial buckling damage. This scenario simulates a case where the underground pipe undergoes significant compressive loading that causes buckling, is subsequently restored by internal pressure, and

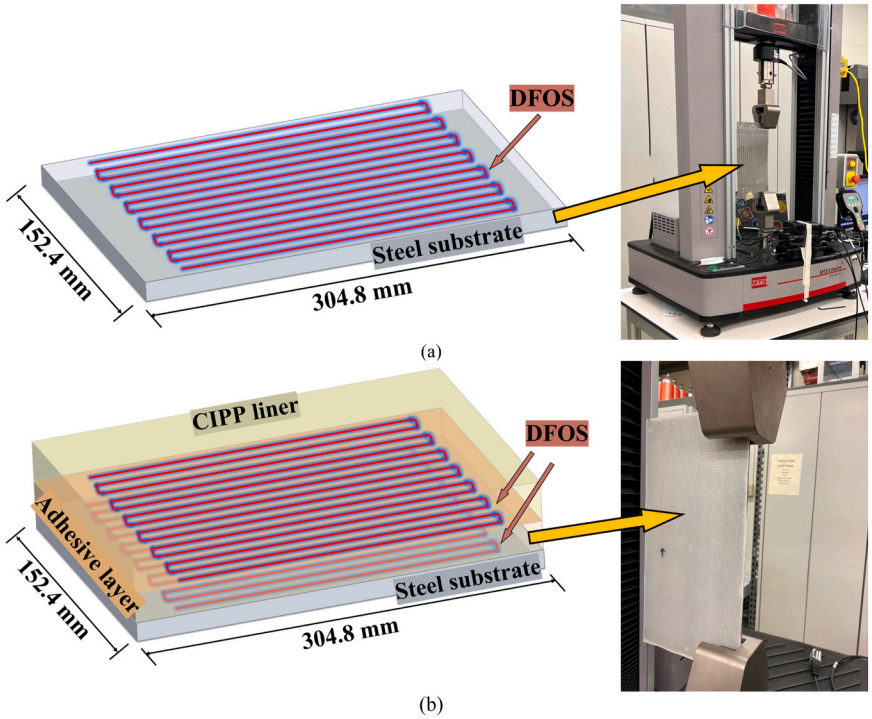


Fig. 2. (a) The schematic of single steel substrate sample and eccentric compressive test setup. (b) The schematic of liner-rehabilitated steel substrate and eccentric compressive test setup.

then experiences a second buckling at the same location.

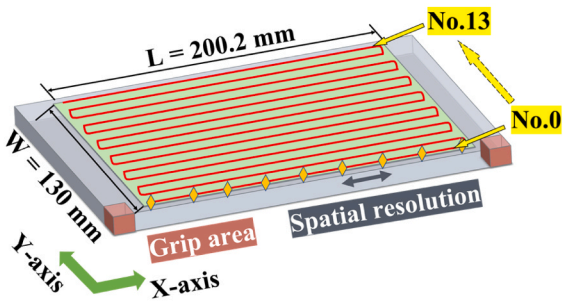
Flat steel substrate samples are used in this study to provide a controlled testbed to validate the workflow before transferring the framework to real pipes. The eccentric compression–induced buckling mode examined here is representative of local wall instability observed in pipelines, and the planar geometry ensures well-controlled boundary conditions and high-quality strain data acquisition. The zig-zag DFOS layout is adopted to establish dense strain data collection and is not intended to reflect field installation practices. In real applications, DFOS would be installed along axial paths with sparser spacing, and the proposed workflow remains compatible with such layouts. Although the interrogator used here has a limited sensing length, the framework does not depend on this specific hardware. Commercially available DFOS systems with sensing ranges up to 100 m can support the proposed framework in instrumented segment-level monitoring in CIPP-rehabilitated pipeline systems.

2.2. Buckling strain field generation from distributed fiber optic sensing

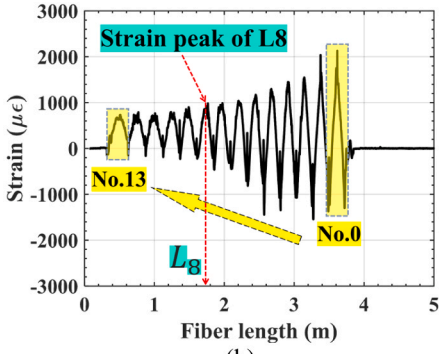
The measurement performance of DFOS is highly dependent on interrogator settings, including sampling rate and SR. An excessively high sampling rate can generate redundant data, not only occupying excessive storage and computational resources but also containing unnecessary noise. In previous engineering applications, the DFOS system typically operated at sample rates ranging from 2 to 100 Hz [40,41]. Therefore, a sampling rate of 10 Hz is selected in this study to balance data accuracy and measurement efficiency for validating the framework under controlled laboratory conditions. In real-world pipeline monitoring, the sampling rate can be flexibly adjusted according to operational requirements. For example, higher rates may be used during hazard events or for critical segments, and lower rates are sufficient for routine long-term SHM. SR, also referred to as gauge pitch, defines the fixed distance between consecutive measurement points along the optical fiber. Given the relatively small measurement area in this study, as shown in Fig. 3, a large SR is insufficient for capturing fine strain gradients. Therefore, the smallest available SR of 0.65 mm is selected to achieve a dense strain measurement across the measurement area.

Fig. 3 illustrates the procedures for converting one-dimensional DFOS strain data into a two-dimensional strain field for the test sample. As illustrated in Fig. 3(a), the DFOS is installed at the center of the sample surface in a zig-zag pattern, forming 14 distinct measurement lines, each with a uniform length of 200.2 mm and a spacing of 10 mm. No.13 denotes the first line at the far end, and No.0 is the last line positioned nearest to the grip area, which experiences the largest buckling deformation. The orange diamonds represent the individual measurement points along the fiber. As previously discussed, the distance between two contiguous measurement points corresponds to the SR of 0.65 mm. Overall, the DFOS covers an area of approximately 130 mm × 200.2 mm for strain monitoring purposes. The raw DFOS data, continuously collected and transmitted to the interrogator, is visualized in real-time on the control platform. However, the raw data is continuous in one dimension, a method is required to distinguish the strain results along each of the 14 measurement lines distributed across the sample.

To achieve this, the strain distribution along the fiber length is plotted at a specific time, as shown in Fig. 3(b). It can be observed that 14 strain peaks gradually increase along the fiber length at intervals of approximately 210.2 mm (measurement line length + spacing), corresponding to No.13 to No.0, getting close to the grips. The results show a promising measurement performance, as each strain peak exhibits a symmetric response, indicating that the strain distribution is symmetric with respect to the central transverse plane, which matches the experimental expectation. By locating each strain peak, the X-Y coordinates of the midpoint of each line are determined, as illustrated in Eq. (1). Based on each midpoint, the X-Y coordinates of remaining measurement points are then calculated according to. Because of the zig-zag installation pattern, the directions of the even-numbered measurement lines are aligned with the positive X-axis, while the odd-numbered measurement



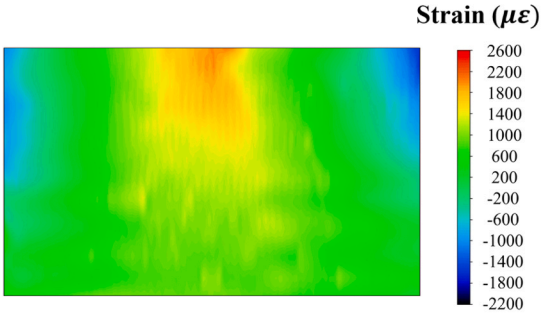
(a)



(b)

$\varepsilon(x,y)$		x				
		0	…	154	…	308
y	0	-1368.5	…	2130.4	…	-1343.2
	1	-1544	…	2036.4	…	-1345.9
	…	…	…	…	…	…
	13	-10.9	…	700.6	…	-26.9

(c)



(d)

Fig. 3. The illustration of the procedures to convert DFOS-measured 1D strain data to a 2D strain field. (a) Install the DFOS in a zig-zag pattern; (b) Plot DFOS strain data along the fiber length and identify the 14 measurement lines; (c) Organize DFOS strain data based on measurement points along the measurement lines; (d) Convert 1D DFOS strain data to a 2D strain field.

lines are oriented in the opposite direction; therefore, Eq. (2) is provided for coordinate conversion along the entire fiber length.

$$\varepsilon\left(\frac{L}{2\times SR},y\right)=\varepsilon(L_y) \tag{1}$$

$$\varepsilon(x,y)=\begin{cases}\varepsilon\left(L_y-\frac{L}{2}+x\times SR\right), \textit{ when } y \textit{ is even}\\ \varepsilon\left(L_y+\frac{L}{2}-x\times SR\right), \textit{ when } y \textit{ is odd}\end{cases} \tag{2}$$

where x is the normalized x-coordinate of each measurement point, ranging from 0 to $\frac{L}{SR}$ at an interval of 1. y is the normalized y-coordinate of each measurement point, which is an integer from 0 to 13 in this study, representing the 14 distinct measurement lines. L is the length of the measurement area (200.2 mm), and L_y is the fiber length corresponding to the strain peak of measurement line y.

Following the coordinate conversion method, the one-dimensional strain data is converted into two-dimensional strain data in the X–Y coordinate system. As illustrated in Fig. 3(c), in this conversion, the raw DFOS data is organized into 14 rows, each representing a measurement line, with strain values uniformly spaced at each measurement point. Because DFOS records strain at discrete positions, linear interpolation is applied between neighboring points to reconstruct a continuous strain profile across the monitored area. This table is updated every 0.1 s, forming a real-time, continuous strain monitoring with 4326 measurement points. Afterward, by mapping the two-dimensional strain data onto their corresponding coordinates, a DFOS-measured strain field is generated, as shown in Fig. 3(d), enabling real-time visualization of the strain distribution across the monitoring area. This conversion facilitates clear visual interpretation of strain distribution and supports DL-based strain field correlation with the FEA results.

2.3. Development of finite element analysis-based virtual testing for buckling behavior

To simulate the mechanical behavior of the test samples under eccentric compressive load, FEA is performed using ABAQUS/CAE 2024. This simulation-based approach enables accurate prediction of structural responses and reduces the reliance on physical testing, offering both cost-effectiveness and deeper insights into mechanical behavior [42]. Two distinct finite element models are developed to represent the experimental conditions. Fig. 4 exhibits the finite element models of the single steel substrate and the CIPP-liner rehabilitated steel substrate. Due to the simple single-layer geometry and thin cross-section of the single steel substrate, this sample is modeled using shell elements, specifically 4-node doubly curved shell elements with reduced integration (S4R). The developed finite element model simulates the CIPP liner-rehabilitated steel substrate, which includes three layers: the steel substrate, an adhesive layer, and a fiber-reinforced polymer CIPP liner. This multilayered system is modeled using solid elements to provide a more accurate representation of the interfacial interactions and surface strain development. The 8-node linear brick element with improved surface stress visualization (C3D8S) is selected for discretization, owing to its enhanced capacity in estimating the surface strain. The material properties of the two samples are presented in Table 1. Since steel exhibits significant ductility after yielding, showing a smooth transition from elasticity to plasticity, an appropriate material model is essential to characterize its nonlinear performance under buckling conditions. The Ramberg-Osgood model is a theoretical mechanics model that describes the nonlinear stress-strain relationship of a material around the yield point [43]. It is popularly employed for the metallic materials that harden with plastic deformation, such as steels [44]. Therefore, the steel substrate is modeled using the Ramberg-Osgood constitutive relationship, as shown in Eq. (3), to simulate its performance of strain hardening and high ductility.

$$\varepsilon=\frac{\sigma}{E}+\alpha\frac{\sigma}{E}\left(\frac{\sigma}{\sigma_Y}\right)^{n-1} \tag{3}$$

where ε is the strain, σ is the stress, σ_Y is the yield stress, E is Young's modulus, α and n are constants that depend on material. In this study, the α and n are assumed to be 1.72 and 12, respectively, according to [45]. The adhesive [46] and CIPP liner [37] have significantly lower stiffness compared to the steel substrate and behave elastically; therefore, they are modeled using linear elastic properties [47]. The interfacial property is simplified as the tied constraint, given the robust bonding achieved by the adhesive layer. Boundary conditions are applied to reflect the experimental loading setup. The bottom corner of the model is fixed to prevent both translation and rotation. The top corner is also constrained against rotation and allowed to move only vertically by 7 mm, in alignment with the experimental eccentric compressive test setup. These constraints ensured accurate reproduction of the loading path and boundary influence observed in the laboratory. Lastly, a mesh sensitivity study is conducted on the single steel substrate to determine the appropriate mesh size for discretization. Several mesh sizes are tested, and the simulation results are compared to assess convergence behavior. According to the results shown in Fig. 5, the

Table 1
Material properties applied in the finite element models.

Sample	Material	Material properties
Single steel substrate	Steel	Young's modulus = 210 GPa Poisson's ratio = 0.29 Yield strength = 350 MPa Tensile strength = 460 MPa
Liner-rehabilitated steel substrate	Steel	Young's modulus = 210 GPa Poisson's ratio = 0.29 Yield strength = 350 MPa Tensile strength = 460 MPa
	Adhesive	Young's modulus = 1 GPa Poisson's ratio = 0.3
	CIPP liner	Young's modulus = 400 MPa Poisson's ratio = 0.4

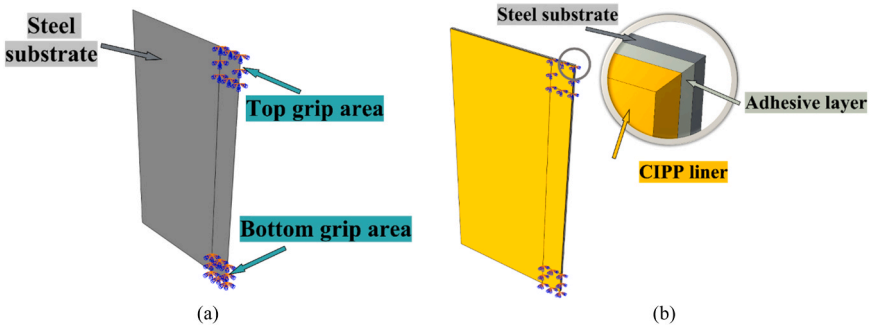


Fig. 4. The finite element models of (a) single steel substrate and (b) liner-rehabilitated steel substrate.

finite element model with a mesh size of 1.25 mm exhibits a negligible difference in buckling load compared to the model with a smaller mesh size. Therefore, a mesh size of 1.25 mm is selected uniformly for this study to balance computational accuracy and efficiency.

The buckling analysis in this study is performed through a two-stage simulation approach, combining both linear and nonlinear methods to comprehensively capture the structural behavior under eccentric loading. The first stage involves a linear eigenvalue buckling analysis, which provides an estimate of the critical buckling load and identifies the associated buckling mode shapes. This analysis is based on the assumptions of small deformations and linear elastic material properties, making it computationally efficient. However, it does not account for real-world effects such as material plasticity, geometric imperfections, or post-buckling response, which can significantly affect the actual structural performance. To overcome these limitations and obtain a more realistic assessment, a nonlinear buckling analysis is carried out using the Riks method. This advanced technique accommodates both geometric and material nonlinearities and enables a way of tracing the entire load–displacement path, including post-buckling behavior. Nonlinear material properties are specified in the model, and geometric nonlinearity is activated to simulate large deformation accurately. Initial imperfections are introduced into the model to reflect the structural irregularities that occur in practice. These imperfections are derived from the first three buckling mode shapes identified in the linear eigenvalue analysis. The amplitude of the first mode shape is varied in a parametric sensitivity study, ranging from 1 % to 50 % of the total sample thickness, to investigate its influence on the buckling response [48]. The second and third mode shapes are scaled to half the amplitude of the preceding mode, respectively, following common practices in imperfection modeling [49]. Once the nonlinear Riks analysis is complete, the predicted strain distribution is compared with the DFOS strain field for validation purposes. The validated strain results are then mapped to corresponding deformation shapes, which form the basis of the digital twin framework. This integrated approach enables real-time visualization and interpretation of complex buckling deformation in the underground pipeline structure.

2.4. Buckling deformation reconstruction through deep learning-based strain field correlation

Minor discrepancies between experimental and numerical strain results may occur due to differences in measurement resolution and data interpolation, as experimental DFOS strain fields are obtained at discrete measurement points, while numerical results represent a continuous strain field. To bridge this gap, this research employs a DL-based convolutional neural network (CNN) to correlate strain fields and identify the most similar pair between the DFOS-measured and FEA-predicted strain fields. By correlating DFOS-measured strain fields with FEA predictions in this study, a digital twin framework can be developed for underground pipelines. As illustrated in Fig. 6, the process begins with organizing DFOS and FEA strain fields into image formats, which are then input into a CNN model for feature extraction. Cosine similarity is computed between the extracted feature vectors to quantify the similarity between each pair of strain fields. By iterating through the entire FEA dataset, the pair with the highest similarity to the DFOS result is selected. To improve the robustness and accuracy of feature extraction, a pre-trained ResNet18 model is employed. ResNet, a widely recognized CNN model, is known for its strong performance in feature extraction across various engineering applications [50]. ResNet18, one of the smallest models of ResNet, is employed for its balance between computational efficiency and extraction capacity. To ensure reproducibility, the pretrained ResNet18 weights provided by PyTorch 2.6.0 and trained on the ImageNet-1K dataset [51] are utilized to build the data-matching method. ResNet18 processes the transformed strain field images and outputs a feature map, which is subsequently flattened into a one-dimensional vector for similarity evaluation. Once the best match is identified, the corresponding FEA deformation result is used to develop and update the digital twin framework, enabling a real-time visualization of the structural behavior.

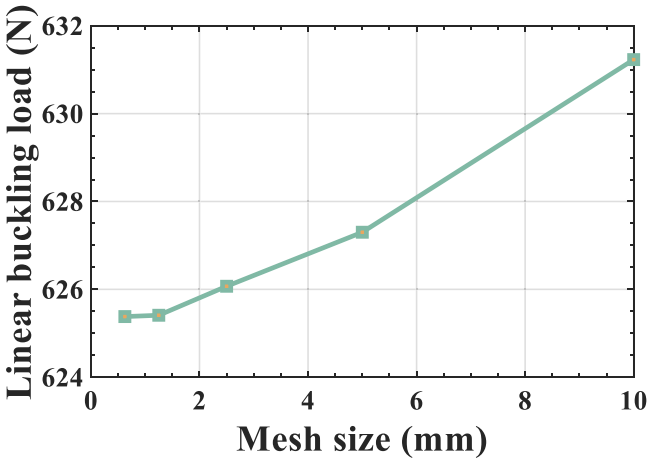


Fig. 5. Mesh sensitivity study results.

2.5. Damage identification and localization: plastic deformation and adhesive crack

Real-time damage detection in the underground pipeline infrastructure using DFOS requires automated approaches for identifying and localizing structural irregularities. This section introduces two approaches for damage identification and localization: one based on detecting peaks in strain rate to indicate crack initiation, and the other based on identifying plastic regions of deformation in terms of strain distributions. A distinct peak in strain distribution consistently occurs when a crack intersects the DFOS pathway, providing a reliable signal for identifying its location [52]. To detect these peaks accurately, a finer SR of strain data is needed. This requires interpolation techniques to estimate strain at points between the discrete measurement points imposed by the fixed SR of DFOS. Two commonly adopted interpolation strategies in two-dimensional domains are bilinear interpolation and triangular interpolation, both of which are useful for reconstructing continuous data surfaces from sparse measurements [53]. Bilinear interpolation is a method that uses repeated linear interpolation to interpolate functions of two variables, typically used on regular grids. Given the uniform SR of DFOS, the collected strain data is uniformly distributed. Therefore, bilinear interpolation is employed to estimate the strain on a finer grid in this study.

Bilinear interpolation approximates the strain $\varepsilon(x,y)$ at a given point (x,y) within a square cell formed by four neighboring nodes P_1, P_2, P_3, P_4, using Eq. (4).

$$\varepsilon(x,y) \approx \alpha_1 \bullet \varepsilon_1 + \alpha_2 \bullet \varepsilon_2 + \alpha_3 \bullet \varepsilon_3 + \alpha_4 \bullet \varepsilon_4 \tag{4}$$

where ε_i represents the strain at point P_i, and the weight α_i are computed based on the relative distances from the interpolation point, as follows:

$$\alpha_1 = \frac{(x_2 - x)(y_2 - y)}{(x_2 - x_1)(y_2 - y_1)}$$

$$\alpha_2 = \frac{(x_2 - x)(y - y_1)}{(x_2 - x_1)(y_2 - y_1)}$$

$$\alpha_3 = \frac{(x - x_1)(y_2 - y)}{(x_2 - x_1)(y_2 - y_1)}$$

$$\alpha_4 = \frac{(x - x_1)(y - y_1)}{(x_2 - x_1)(y_2 - y_1)}$$

where $(x_1, \ y_1)$, $(x_1, \ y_2)$, $(x_2, \ y_1)$,$(x_2, \ y_2)$ are the coordinates of

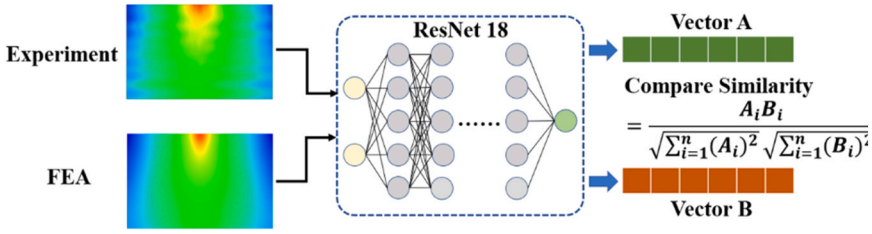


Fig. 6. The illustration of DL-based strain field correlation.

the four neighboring lattice points P_1, P_2, P_3, P_4.

To analyze strain gradients, a numerical approximation of the derivative is given in Eq. (5).

$$\nabla\varepsilon(x) \approx \frac{\varepsilon(x+\Delta x)-\varepsilon(x)}{\Delta x} \tag{5}$$

As $\Delta x \to 0$, the approximation converges to the exact gradient. A denser grid provides a gradient closer to the true value but increases computational costs. A grid resolution of 0.1 mm × 0.1 mm is selected based on a sensitivity study, as further refinement shows little improvement in estimation accuracy. The identified strain rate peak will be incorporated into the digital twin model to present the crack location and severity.

Furthermore, steel material exhibits strain-hardening behavior and a loss-of-ductility stage beyond yielding. Given the strain results provided by DFOS, the plastic region of deformation on the single steel substrate can be readily identified. Although two DFOS are embedded into the liner-rehabilitated steel substrate sample, due to cost and installation difficulties, integrating DFOS within the liner is more practical, as the installation of DFOS on the buried underground pipe can be significantly challenging. To address this, a theoretical analysis is performed using the strain results collected from the adhesive layer-liner interface (green dot) to estimate the strain results at the adhesive layer-steel interface (red dot), thereby further predicting the plastic region of deformation on the steel substrate, as shown in Fig. 7.

Since the three layers are perfectly bonded, strain is continuous across the interfaces. Therefore, a linear strain distribution through the thickness of the liner-rehabilitated steel substrate is assumed. Due to the significant difference in Young's moduli between the steel and adhesive, the neutral axis is not located at the geometric midplane. To verify this, the position of the neutral axis is computed as detailed below.

The stress in the adhesive layer (σ_0) and steel (σ_1) can be calculated using Eqs. (6) - (7):

$$\sigma_0 = E_0(\varepsilon_0 + \kappa z) \quad (0 \le z \le h_0) \tag{6}$$

$$\sigma_1 = E_1(\varepsilon_0 + \kappa z) \quad (h_0 \le z \le h_0 + h_1) \tag{7}$$

where E_0 is the Young's modulus of adhesive, E_1 is the Young's modulus of steel, ε_0 is the given strain results at the adhesive layer, κ is the curvature, and z_n is the neutral axis position.

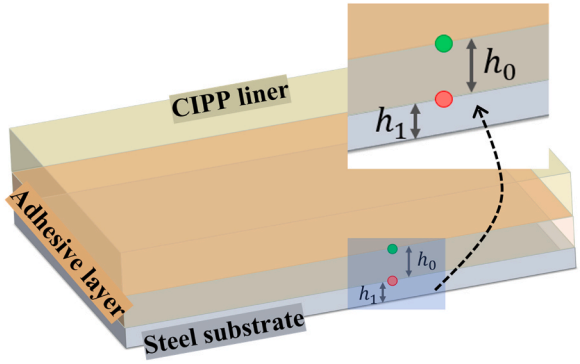


Fig. 7. Schematic of the liner-rehabilitated steel substrate.

The resultant force is calculated by Eq. (8):

$$\int_0^{h_0} \sigma_0 dA + \int_{h_0}^{h_0+h_1} \sigma_1 dA = 0 \tag{8}$$

Assuming a unit width ($dA = dz \bullet 1$), Eq. (8) can be rewritten as Eq. (9):

$$E_0 \int_0^{h_0} [\varepsilon_0 + \kappa(z - z_n)]dz + E_1 \int_{h_0}^{h_0+h_1} [\varepsilon_0 + \kappa(z - z_n)]dz = 0 \tag{9}$$

For this scenario, the $\varepsilon_0 = -\kappa z_n$, strain is zero at $z = z_n$, The neutral axis is positioned at the centroid of the transformed section. Eq. (9) is then simplified to Eq. (10).

$$z_n = \frac{E_0 \bullet \frac{h_0^2}{2} + E_1 \bullet h_1 \bullet (h_0 + \frac{h_1}{2})}{E_0 h_0 + E_1 h_1} \tag{10}$$

Therefore, the strain in the steel layer can be estimated from the strain in the adhesive layer, as provided in Eq. (11):

$$\varepsilon_1 = \varepsilon_0 + \kappa h_0 \tag{11}$$

The estimated strain results will be validated first and then incorporated into digital twin model to identify plastic region of deformation.

3. Results and discussion

3.1. Damage identification from the digital twin of steel substrate

3.1.1. Strain field demonstration from distributed fiber optic sensor and virtual testing

To enable effective visualization and facilitate comparison, the one-dimensional DFOS strain data collected along the steel substrate surface are converted into a two-dimensional strain field, using the method outlined in Section 2.2. Fig. 8 presents the strain field evolution as vertical displacement increases during loading. Positive tensile strain appears near the midpoint of the edge of the grip areas, indicated by red regions, and negative compressive strain develops at the side of the grip areas, highlighted in blue. Overall, the strain field demonstrates an approximately symmetric pattern with respect to the centerline between the two grip areas. This symmetry aligns with the expected structural response, given the symmetric positioning of the grip areas on the steel substrate. Minor deviations from perfect symmetry are observed and likely caused by instrumental limitations. Despite the sample being securely gripped, minor shifts in the experimental setup may introduce minor inconsistencies in boundary conditions, resulting in asymmetric strain responses. Due to the advantage of numerical analysis being free of external disturbances, the comparison results show that FEA has a symmetric strain distribution, which is highly similar to the results from DFOS.

To validate the feasibility of the finite element model in virtually testing the single steel substrate subjected to eccentric buckling, the load-displacement curve is compared, and the differences are quantified in strain fields between the experiment and simulation. Fig. 9 shows the load-displacement curves obtained from experimental and FEA results.

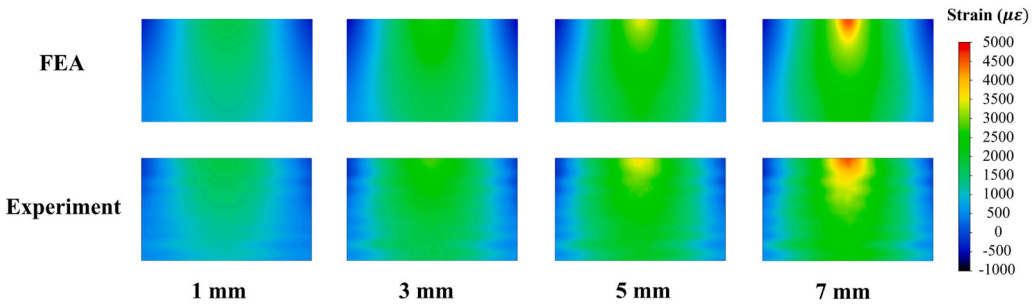


Fig. 8. Comparison of strain fields of single steel substrate at displacement levels of 1 mm, 3 mm, 5 mm, and 7 mm.

It is observed that the buckling loads are greatly matched, except for the finite element model with a 50 % initial imperfection. The difference in buckling load between the experiment and the simulation with a 1 % initial imperfection is 0.0044 %. As the initial imperfection increases, the buckling load and computational time decrease. Although the model with a 1 % initial imperfection shows the closest buckling load, it shows a slight discrepancy in the pre-buckling stage and requires more computational time. As shown in the zoom-in view in Fig. 9, the model with a 25 % initial imperfection aligns well with the experimental results, with only a 1.9 % difference in buckling load. It reduces the computational time to 1/4 of that required for the 1 % initial imperfection case. Therefore, the model with a 25 % initial imperfection is selected for the following study. In the post-buckling stage, as the displacement increases, the load on the sample decreases, and the FEA-predicted load also exhibits a great agreement with the experimental results. Overall, the FEA results match well with the experimental data, indicating the accuracy of the finite element model. For further validation, a pixel-based comparison method is applied to quantify the similarity between DFOS and FEA results, using the parameters including structural similarity index measure (SSIM) and mean absolute percentage error (MAPE). As shown in Fig. 10 (a). The SSIM values for over 300 pairs of strain fields from 0 to 7 mm vertical displacement are above 0.89, indicating that the strain results from DFOS and FEA have high similarity. Additionally, MAPE results, presented in Fig. 10 (b), provide the relative error percentage between DFOS and FEA fields, demonstrating that the difference remains below 4.1 %.

In addition to the initial 7 mm vertical displacement that induced eccentric buckling, the steel substrate is restored to its original shape and subjected to another 4 mm vertical displacement to experimentally investigate the structural responses after buckling damage. The buckling behavior of the single steel substrate under the second-round eccentric load is also analyzed numerically. Fig. 11 presents the comparison of the load-displacement curves from the second eccentric compressive test. Different from the initial test, the steel substrate is already in a buckled state during the second test. As a result, the compressive load increases with the vertical displacement, while the rate of load increase decreases. The FEA results exhibit a similar trend to the experimental results, demonstrating the capability of FEA in capturing the entire buckling mechanical performance of the steel substrate. Overall, the comparison demonstrates that the FEA results are in good agreement with the DFOS results, validating the accuracy of the finite element model. Building on this validation, FEA can complement DFOS by extending the capacity from strain monitoring to comprehensive mechanical performance monitoring and facilitating the development of the digital twin framework.

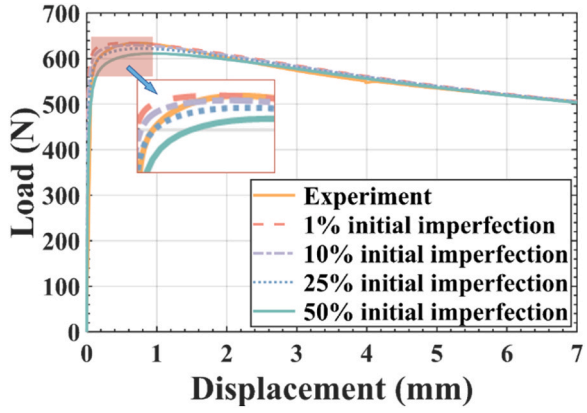


Fig. 9. Comparison of load-displacement curves for eccentric compressive test.

3.1.2. Buckling deformation reconstruction and plastic regions of deformation visualization

To reconstruct the shape of buckling-deformed single steel substrate for the development of the digital twin framework, the strain field correlation is essential for an accurate representation of deformation. In terms of the strain field correlation method introduced in Section 2.4, the DFOS strain fields are input into the CNN model and set as the ground truth. The FEA strain fields are organized into a database, which includes all FEA strain fields at the displacement levels from 0 to 7 mm. By extracting the features from DFOS strain fields and iteratively comparing the cosine similarities within the FEA strain field database, the most similar pairs are identified. Fig. 12 shows the results of strain field correlation. The corresponding displacements of the matched strain field pairs are compared, showing an average displacement difference of 0.754 mm. This difference is minor and remains within an acceptable sub-millimeter-level range, making it suitable for practical applications in the underground pipeline industry [54].

Based on strain correlation results, this study reconstructs the deformation and develops the digital twin framework of the buckled single steel substrate. Fig. 13 (a) shows the 3D digital twin framework for visualizing the deformation of the single steel substrate subjected to eccentric compression. The left figure illustrates that as the vertical displacement of the top grip increases to 7 mm, the buckling-induced deformation of the steel substrate progressively intensifies. The four figures on the right provide separate visual representations of the deformation at different displacement levels, further demonstrating that the highest deformation is consistently observed at the center of the free edge between the grips. Additionally, for typical structural steel, the yield strain is commonly around 0.002 [37]. When the estimated strain in the steel substrate exceeds the yield strain, the corresponding region is classified as the plastic region. Fig. 13 (b) illustrates the developed 3D digital-twin framework for visualizing plastic regions of deformation in the single steel substrate. These regions originate at the edges of grip areas and expand to the high deformation regions as eccentric compression increases. The above method leverages the continuous measurement capacity of DFOS, enabling the detection and mapping of the potential damage. It enhances the efficiency of real-time structural damage monitoring and can be employed to identify potential damages in liner-rehabilitated underground pipeline infrastructure, ultimately extending the structural service life.

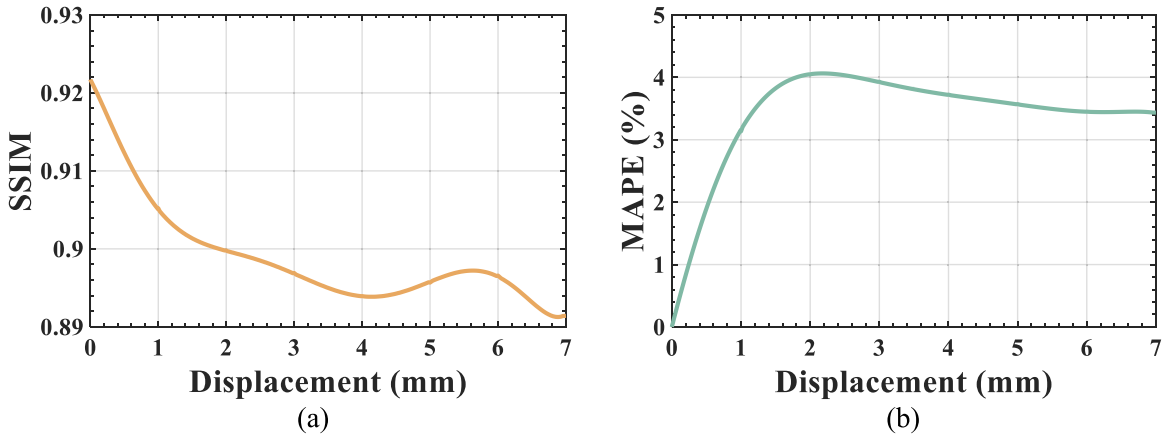


Fig. 10. Quantification of difference between experiment and FEA results. (a) SSIM; (b) MAPE.

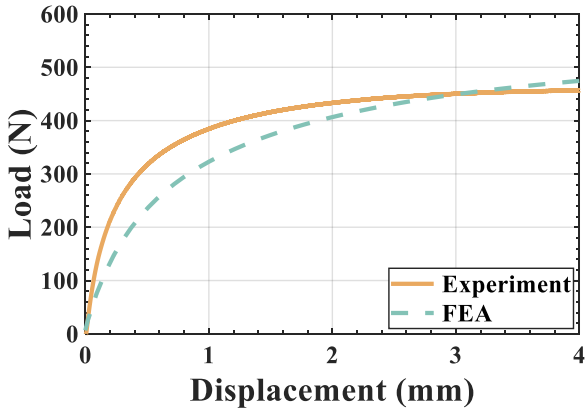


Fig. 11. Comparison of load-displacement curves for the second eccentric compressive test.

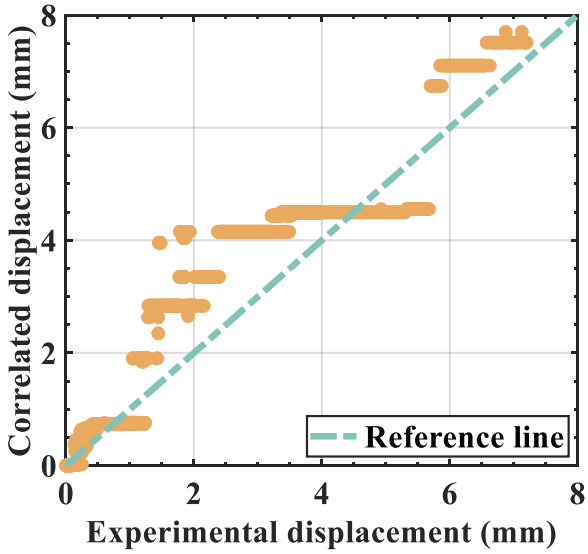


Fig. 12. The DL-based strain field correlation results at distinct displacement levels.

The digital twin framework developed in this section captures the key structural deformation and plastic deformation characteristics, providing a predictive framework for real-time monitoring. The comparison and correlation between the experimental and numerical strain fields verify the accuracy and reliability of the digital twin framework. This framework enables continuous tracking of deformation states, which is crucial for real-time SHM, enabling a dynamic, real-time virtual representation of the single steel substrate. This study highlights the potential of developing digital twin frameworks for self-sensed underground pipelines.

3.2. Damage identification from the digital twin of liner-rehabilitated steel substrate

Different from the single substrate, the liner-rehabilitated steel substrate has two additional layers: CIPP liner and adhesive layer. The liner is attached to the steel substrate through an adhesive layer, simulating the application of a CIPP liner in rehabilitating an underground pipeline. However, damages on the adhesive layer, such as cracks, can lead to poor bonding between the liner and the pipe, compromising the rehabilitation performance of the CIPP liner [55]. In addition, the plastic deformation that occurs on the underground pipe poses a significant risk, as it is an irreversible damage. To date, the plastic deformation on the pipe has not been systematically investigated through SHM approaches [56]. To investigate both adhesive crack and plastic deformation damages, a DFOS-integrated liner is employed. Although the DFOS is installed directly on the steel substrate, the installation of DFOS on an underground pipeline presents a practical challenge. Therefore, the strain measurement results obtained from the DFOS-integrated liner are used to identify and localize the cracks and plastic deformation on the liner-rehabilitated steel substrate. The findings are validated using measurements from the DFOS on the steel substrate and through FEA, allowing us to assess the effectiveness and accuracy of the DFOS-integrated liner in damage detection. Fig. 14 shows the procedures of the study on the liner-rehabilitated steel substrate.

3.2.1. Strain field demonstration from distributed fiber optic sensor and virtual testing

The accuracy of the finite element model is demonstrated in Section 3.1.1 through the comparisons of strain fields, load-displacement curve, SSIM, and MAPE results between experimental and numerical results. Building upon the previous findings, this section focuses on the strain fields at two critical interfaces: the adhesive layer-steel substrate interface and the adhesive layer-liner interface. Compared to newly constructed pipes, the developed digital twin framework is especially valuable for existing in-service pipes, especially those aging and deteriorated pipes. Before installing the CIPP liner on the internal surface of an underground pipe, the integration of DFOS with the CIPP liner enables the function of self-sensing without compromising the rehabilitation performance of the liner. To ensure an effective rehabilitation outcome, the condition of the adhesive layer is a key concern. Therefore,

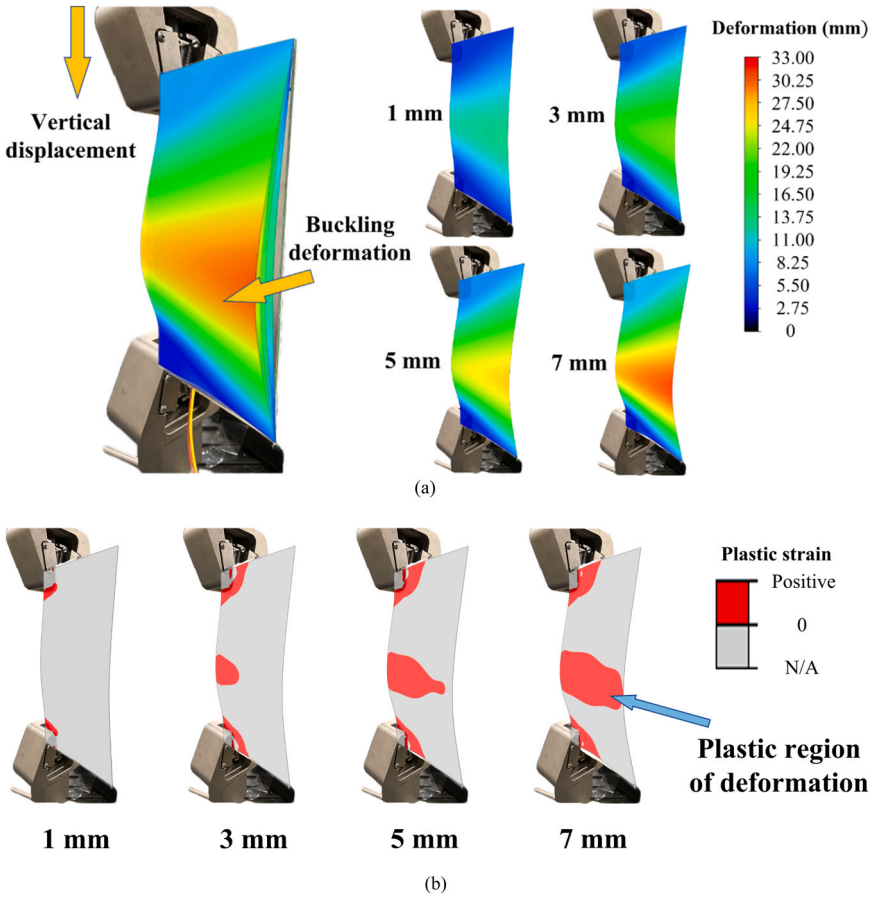


Fig. 13. The developed 3D digital-twin framework for (a) structural deformation and (b) plastic region of deformation visualization of single steel substrate.

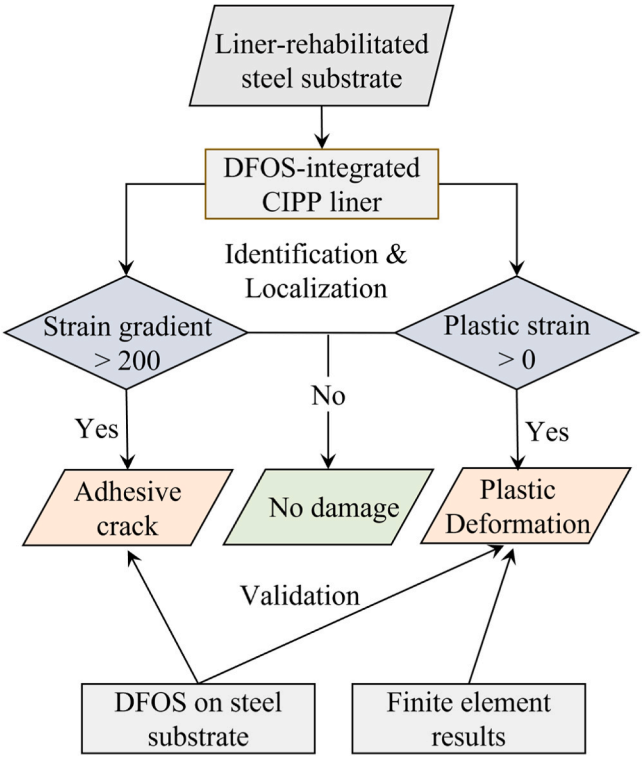


Fig. 14. Procedures to identify damages on liner-rehabilitated steel substrate.

two interfaces are installed with DFOS and presented in this section. The strain fields at these two interfaces are generated at 4 distinct vertical displacement levels, as shown in Fig. 15. Again, the FEA results match well with the experimental results, reemphasizing the accuracy of the finite element model. Compared to the strain field at the adhesive layer-steel substrate interface, as shown in Fig. 15 (a), the strain field at the adhesive layer-liner interface, as shown in Fig. 15 (b), exhibits significantly larger tensile and compressive strains. The difference is attributed to the thickness of the adhesive layer, with the adhesive layer-liner interface experiencing a greater buckling deformation. Moreover, compared to the strain fields at the adhesive layer-steel substrate interface, those at the adhesive layer-liner interface show a more symmetric and uniform strain distribution. This indicates that integrating DFOS with the CIPP liner results in improved strain monitoring performance, demonstrating the feasibility of the application of DFOS-integrated CIPP liner in the underground pipeline infrastructure.

3.2.2. Buckling deformation reconstruction and adhesive crack visualization

The approach for the structural deformation visualization through the digital twin framework is presented and validated based on the study of a single steel substrate. By correlating the strain fields generated from DFOS and FEA, the mechanical responses of the liner-rehabilitated steel substrate are obtained from the FEA in this section. One difference compared to the single steel substrate sample is that two DFOS are installed at the interfaces of the liner-rehabilitated steel substrate. In this way, two independent strain field correlation analyses are conducted to correlate the results at these two interfaces. If the correlated results differ at the same displacement level, an average correlated displacement is calculated. In this study, the maximum difference in correlated displacement between two interfaces is 0.14 mm. The digital twin

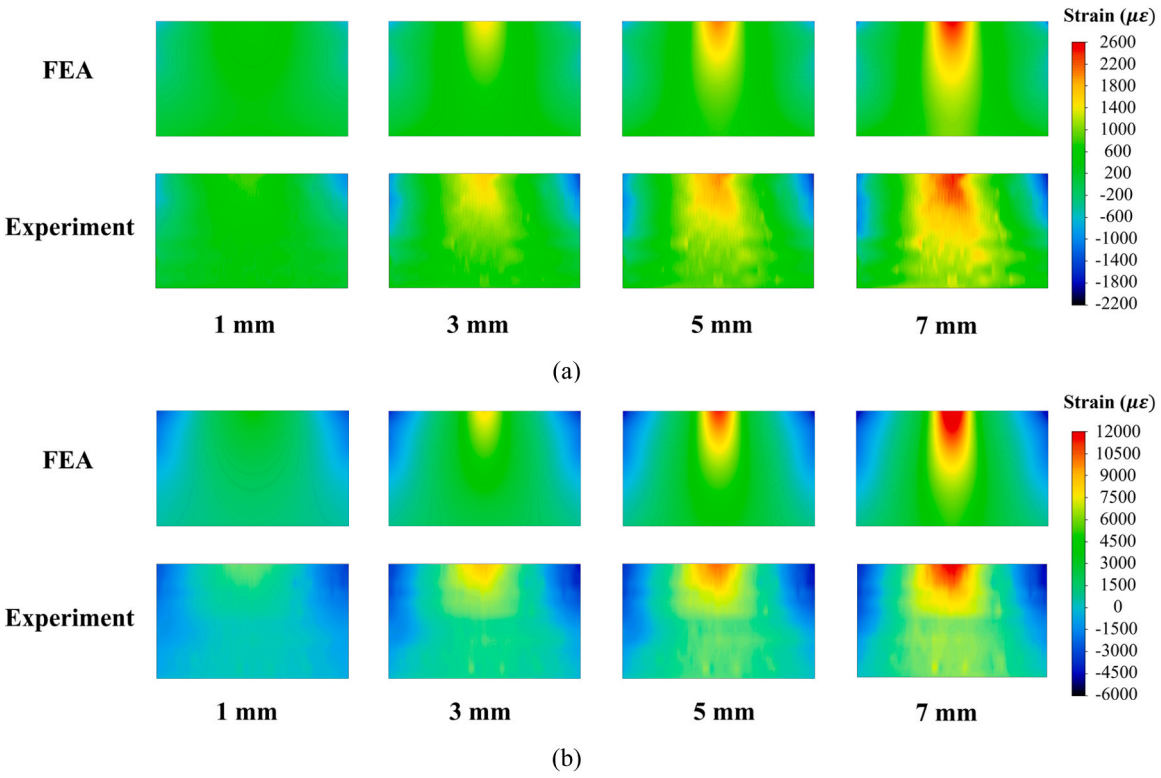


Fig. 15. Comparison of strain fields of (a) the adhesive layer-steel substrate interface, and (b) the adhesive layer-liner interface of liner-rehabilitated steel substrate at displacement levels of 1 mm, 3 mm, 5 mm, and 7 mm.

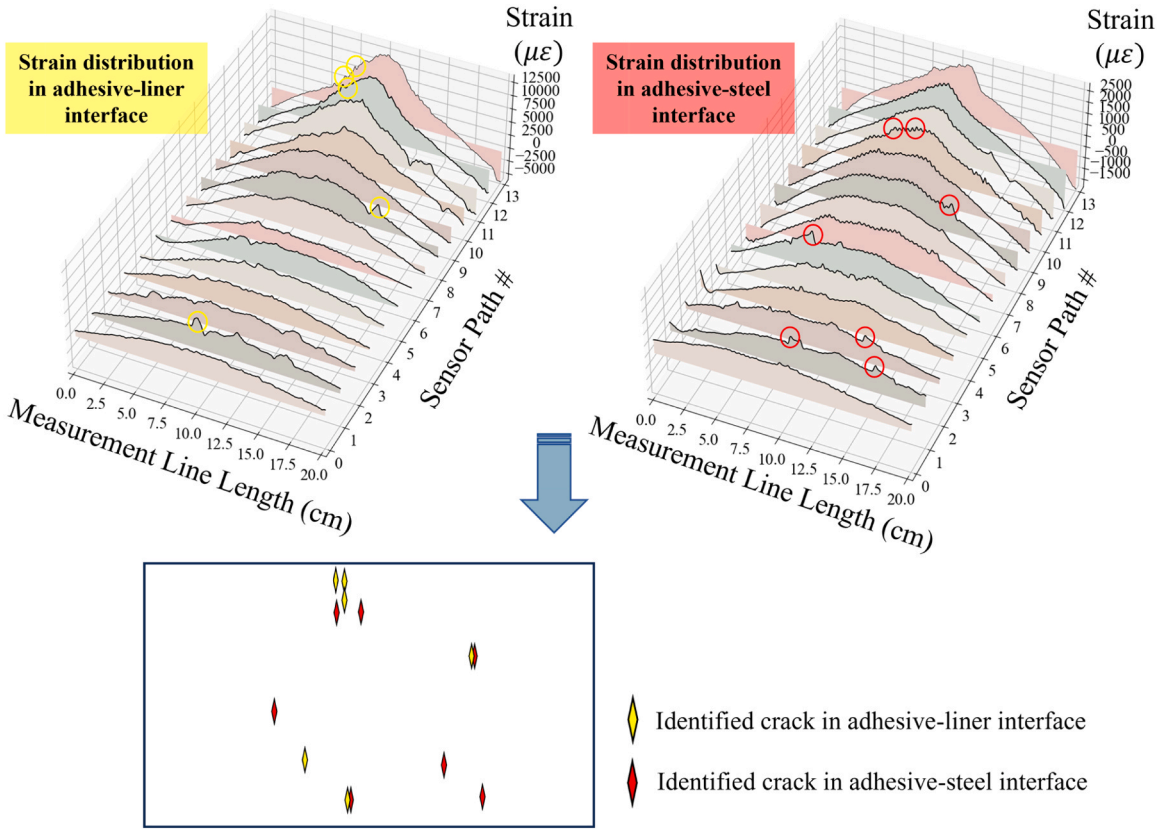


Fig. 16. Interfacial crack identification and localization.

framework presenting the structural deformation of the liner-rehabilitated steel substrate is shown in Fig. 16. The deformed shape and corresponding deformation results are similar to those of the single steel substrate, because the steel substrate is much stiffer than the liner and adhesive layer. Therefore, under the same vertical displacement, the two samples exhibit similar deformation behavior.

The identification of potential damage is important, as it often indicates the locations where failure initiates. To further enrich the content of the digital twin framework, the adhesive crack information is integrated. In the case of the liner-rehabilitated pipe system, crack detection on the adhesive layer is particularly important, as such damage can affect the liner's rehabilitation performance, for example, by causing liner debonding [57]. Furthermore, it is hard to detect cracks on liner-rehabilitated pipe systems visually, underscoring the need for a non-destructive monitoring method. As introduced in Section 2.5, the DFOS strain rate shows a significant increase when it intersects a structural irregularity, such as a crack. By processing the strain data from DFOS using interpolation and gradient analysis, structural damage can be localized by converting the location of the strain rate surge into two-dimensional coordinates. Additionally, with the advantage of the high sensitivity of DFOS, the severity of damage can be assessed using the strain gradient values. Fig. 16 illustrates the interfacial crack identification and localization method. Localized defects appear as very sharp, cone-like features in the strain map, as circled in yellow and red. The gradient highlights these steep spatial changes; therefore, damage areas are those for which the gradient magnitude exceeds a threshold. There is no universal critical strain gradient value at which cracking necessarily occurs [58]. Accordingly, the strain-gradient threshold is progressively varied over a reasonable range, and the value that maximizes overlap with regions exhibiting obviously localized strain concentrations is selected for damage identification. In this study, a strain gradient threshold of 200 is defined; any region with a strain gradient greater than 200 is classified as a crack.

The digital twin framework for liner-rehabilitated steel substrate under buckling is shown in Fig. 17. Red irregular regions represent the cracks identified on the interface between the adhesive layer and steel substrate, and yellow irregular regions indicate the cracks identified on the interface between the adhesive layer and liner. A larger area signifies a more extensive and severe crack. The results show that cracks commonly occur close to the edges, near the high tensile strain region. It is also observed that, although the strain field distribution has a symmetric pattern, the crack distribution is not. This finding underscores the importance of accurate crack identification and localization, as their patterns are unpredictable. Additionally, it is observed that more crack occurs at the adhesive layer–steel substrate interface, highlighting the importance of firm bonding between the liner and the steel substrate in the rehabilitation projects. Although the interface between the adhesive layer and liner has less damage, the damage observed in this interface appears earlier, demonstrating the importance of the DFOS-integrated liner in early damage detection.

3.2.3. *Plastic regions of deformation visualization*

The study on the single steel substrate presents a viable solution to generate the plastic deformation visualization through the developed digital framework. In this section, the plastic deformation visualization is applied to the liner-rehabilitated steel substrate. Due to the cost and installation difficulty, the integration of DFOS and liner is more convenient and practical, as installing the DFOS on the buried underground pipe directly can be significantly challenging. Therefore, the satisfactory performance of DFOS-integrated liners in effectively monitoring the entire pipe condition is important. Given the previous approach, the plastic deformation on the steel substrate can be determined using the strain distribution on the steel surface. As illustrated in Section 2.5, the strain distribution on the adhesive-layer-steel interface is estimated based on the strain results obtained from the adhesive layer-liner interface to investigate if the DFOS-integrated liner is capable of monitoring the strain distribution on the steel substrate. Based on the estimated strain in the steel substrate, the structural condition of the entire substrate can be predicted. Fig. 18 presents the comparison between the estimated plastic regions in the steel substrate obtained from the FEA, experimental measurements, and theoretical predictions. Due to the partial coverage of the substrate by the DFOS, the experimentally measured region appears smaller than those from the FEA and theoretical analysis, both of which show the entire substrate area. The prediction results align well with the FEA and experimental results, with minor differences observed near the edge of the grip areas. Moreover, the development of buckling-induced damages shown in Fig. 17 correlates with the progression of plastic regions of deformation, demonstrating the consistency and reliability of the findings from the developed digital framework. Fig. 19 shows the distribution of plastic regions of deformation in the liner-rehabilitated steel substrate. Consistent with the findings on the single substrate (Fig. 13 (b)), the plastic deformation initiates near the grip areas and propagates toward high deformation regions. Compared to the single steel substrate case,

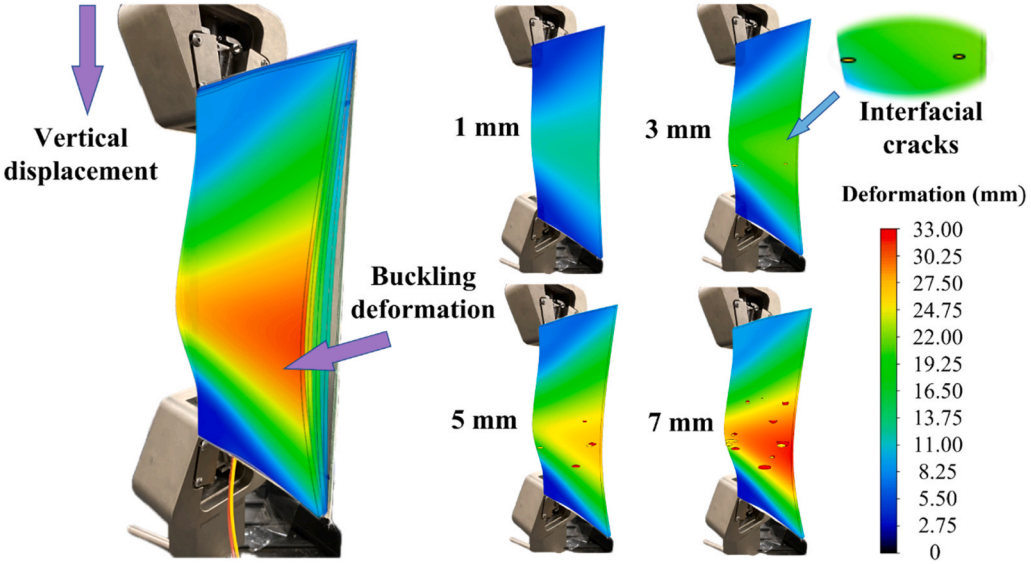


Fig. 17. The developed 3D digital-twin framework for structural deformation and crack visualization of liner-rehabilitated steel substrate.

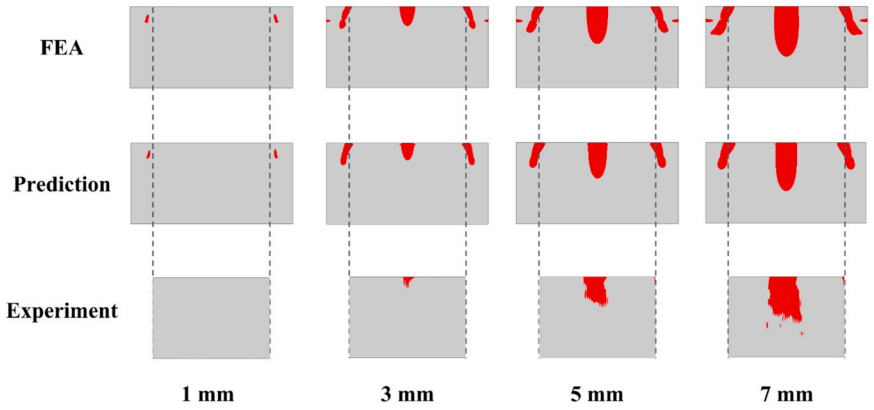


Fig. 18. Comparison of plastic regions of the steel layer of liner-rehabilitated steel substrate at displacement levels of 1 mm, 3 mm, 5 mm, and 7 mm.

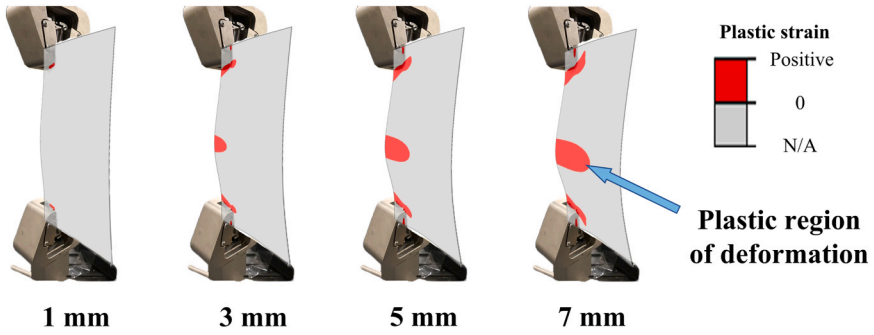


Fig. 19. The developed 3D digital-twin framework for plastic region of deformation visualization of liner-rehabilitated steel substrate.

the plastic regions of deformation in the liner-rehabilitated steel substrate are significantly smaller, indicating the rehabilitation effectiveness of the CIPP liner in protecting the substrate. The results demonstrate that integrating DFOS with the liner enables reliable prediction of the strain-based condition of the steel substrate. This approach shows strong potential for the application of DFOS-integrated liner in the underground pipe industry for SHM purposes.

The developed digital twin framework provides a comprehensive SHM of the liner-rehabilitated steel substrate, incorporating both mechanical responses and physical damage information. The mechanical response, such as the structural deformation performance illustrated in this study, offers a virtual representation of the buckling deformation evolution of liner-rehabilitated steel substrate under eccentric compression. The plastic regions of deformation on the steel substrate are identified and localized, as well as the cracks on the two sides of the adhesive layer. The DFOS-integrated liner demonstrates the capacity to accurately capture the strain and damage information and correlate them with FEA results in developing the digital twin framework. The findings support the feasibility of employing digital twin frameworks for real-time SHM and predictive maintenance in the underground pipeline industry.

4. Conclusion and future work

This research presents a comprehensive framework to develop an enhanced digital twin framework for real-time structural health monitoring (SHM) of underground pipelines. The approach comprises the integration of distributed fiber optic sensing technology, finite element analysis (FEA), and deep learning (DL) algorithms. Through the installation of the distributed fiber optic sensor (DFOS) on the steel substrate and cured-in-place pipe (CIPP) liner, this study demonstrates the feasibility and accuracy of DFOS in real-time strain field monitoring for the steel substrate, highlighting its strong potential for application in the underground pipeline industry. The well-validated finite element model allows for the transformation of localized strain data into broader mechanical performance insights, complementing the functionality for DFOS beyond strain sensing. The DL-based correlation analysis further facilitates the development of the digital twin framework capable of providing a high-fidelity, real-time, three-dimensional representation of structural deformation. In addition to visualizing large-scale structural deformation, minor structural damages and invisible plastic regions of deformation are detected, developing a digital twin framework for damage detection and localization in this study. The developed framework shows comprehensive and robust structural monitoring capabilities. Several summarized conclusions are listed as follows:

- The durability of DFOS under large deformation is demonstrated to be promising. Under the extreme eccentric buckling scenario considered in this study, the DFOS consistently delivers accurate and continuous data, facilitating effective strain field monitoring for underground pipeline infrastructure.
- The FEA results exhibit high accuracy. The numerical model with a 25 % initial imperfection not only improves computational efficiency, but also shows strong agreement with the experimental load-displacement curve, with a buckling load difference of 1.9 %. Strain fields comparison yields a lowest SSIM of 0.89, and a highest MAPE of 4.1 %.
- The entire buckling performance is greatly captured by FEA. This validates the reliability of the developed finite element model in predicting structural behavior of the steel substrate under complex loading conditions, supporting its application in developing the digital twin framework.
- The DL-based strain correlation analysis establishes a foundation for the digital twin framework. The correlated results show a maximum displacement difference of less than 1 mm, demonstrating high correlation accuracy.

- The identified and localized plastic regions of deformation and adhesive cracks reveal the underlying damage mechanism of the steel substrate under buckling conditions, supporting early-stage maintenance and rehabilitation. The damage characteristics further contribute to the development of the digital twin framework, which provides a real-time, three-dimensional representation of the structural physical conditions.
- The DFOS-integrated liner system demonstrates the capacity to detect minor structural damages and invisible plastic regions of deformation for the entire liner-rehabilitated steel substrate. The developed digital twin framework advances the SHM of self-sensed cured-in-place underground pipeline projects.

To conclude, this study introduces an innovative framework to develop an enhanced digital twin system for real-time SHM of underground pipelines. This framework serves as an advanced solution to monitor the pipelines in real time, providing continuous insights into pipeline performance and condition. Furthermore, with the damage identification and localization capacities of DFOS, a digital twin framework is developed to provide a comprehensive understanding of not only the structural mechanical responses but also minor damages at early stages.

While this study is limited to evaluating the developed framework on a liner-rehabilitated steel substrate to ensure precise control of boundary conditions and loading configurations, the approach shows strong potential for broader real-world pipeline applications. The designed sensor installation strategy and sensor data conversion method are inherently scalable. For instance, the zig-zag DFOS installation pattern can be adapted to the cylindrical pipelines by aligning the sensor paths along axial directions. Moreover, the coordinate-based strain data conversion method can be readily applied to real pipelines, as it is formulated based on spatial resolution and sensor spacing, both of which are adjustable to accommodate different monitoring requirements. Although this paper demonstrates the applicability of the developed framework under the eccentric compression conditions, the sensing and damage identification workflow are generalizable to more complex loading and environmental scenarios. Given the widespread applications of CIPP liners in the underground pipeline industry, the sensor installation, data conversion, and loading conditions presented in the framework can be scaled up to extensive networks of pipelines for large-area SHM.

Future research will focus on applying and validating this framework at full-scale pipeline samples subjected to realistic compression and operational loading conditions. In summary, the developed framework addresses a critical gap in current SHM practices for underground pipelines by enabling real-time visualization and precise localization of cracks and plastic regions of deformation. This study lays a solid foundation for a scalable, intelligent, and visually interpretable digital twin solution for underground pipeline infrastructure under real-world conditions.

CRediT authorship contribution statement

Junyi Duan: Writing – original draft, Methodology, Investigation, Formal analysis. **Xingyu Wang:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Huaixiao Yan:** Writing – original draft, Validation, Methodology, Investigation, Data curation. **Sike Wang:** Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Ying Huang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Chengcheng Tao:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] Wang J, Liu G, Wang J, Xu X, Shao Y, Zhang Q, et al. Current status, existent problems, and coping strategy of urban drainage pipeline network in China. Environ Sci Pollut Res 2021;28:43035–49. https://doi.org/10.1007/s11356-021-14802-9.
[2] Wasim M, Shoaib S, Mubarak NM, Inamuddin, Asiri AM. Factors influencing corrosion of metal pipes in soils. Environ Chem Lett 2018;16:861–79. https://doi.org/10.1007/s10311-018-0731-x.
[3] Xi D, Lu H, Zou X, Fu Y, Ni H, Li B. Development of trenchless rehabilitation for underground pipelines from an academic perspective. Tunn Undergr Space Technol 2024;144:105515.
[4] Yan H, Duan J, Wang X, Huang Y, Tao C. Integration of Machine Learning and Finite Element Analysis for Assessing the Conditions of Cure-in-Place Pipe Liner Materials for Damaged Pipe Rehabilitation. Pipelines 2025. Tampa, Florida: American Society of Civil Engineers; 2025. p. 517–25. https://doi.org/10.1061/9780784486382.056.
[5] Duan J, Tao C, Wang X, Huang Y. Computational Assessment of Pitting Corrosion in Cast-Iron Pipelines Rehabilitated by Cured-in-Place Pipe Liner. Pipelines 2025. Tampa, Florida: American Society of Civil Engineers; 2025. p. 384–93. https://doi.org/10.1061/9780784486375.042.
[6] Yan X, Wang X, Deng C, Zhao Y, Mei S, Bai J, et al. Testing and analysis of CIPP liner under simulated groundwater pressure. Tunn Undergr Space Technol 2024;152:105903.
[7] Zhao Y, Ma P, Chen Y, Liu K, Yan X, Ma B, et al. Stability assessment of CIPP liner under varied boundary conditions: a theoretical and simulation study. Results Eng 2024;22:102187.
[8] Ding J., Tao C., Duan J., Wang X., Huang Y. A Machine Learning-Based Interpretable Glass-Box Model for Pipeline Failure Mode Prediction Considering Quantitative Coupling Factors 2025:507–516. https://doi.org/10.1061/9780784486382.055.
[9] Psyrras N, Kwon O, Gerasimidis S, Sextos A. Can a buried gas pipeline experience local buckling during earthquake ground shaking? Soil Dyn Earthq Eng 2019;116:511–29.
[10] Yan X, Wang X, Xiang W, Zhao Y, Ma B. Buckling behavior of formed-in-place-pipe (FIPP) liners under groundwater pressure: an experimental investigation for buried municipal pipelines. Tunn Undergr Space Technol 2023;142:105397.
[11] Nie H, Ma W, He X, Huo X, Wang Y, Ren J, et al. Study on buckling deformation mechanism of pipeline crossing road. Eng Fail Anal 2022;139:106412.
[12] Lu C, Ariaratnam ST, Yan X, Ma B, Zhao Y, Xiang W. Buckling behavior of thin-walled stainless-steel lining wrapped in water-supply pipe under negative pressure. Appl Sci 2021;11:6781.
[13] Wang X, Zhao Y, Ariaratnam ST, Yan X. Study on the impact of ovality defect on structural stability of CIPP liner of drainage pipeline. Tunn Undergr Space Technol 2023;140:105338.
[14] Alam S, Matthews J, Sterling R, Allouche E, Selvakumar A, Condit W, et al. Evaluation of testing methods for tracking CIPP liners' life-cycle performance. Cogent Eng 2018;5:1463594. https://doi.org/10.1080/23311916.2018.1463594.
[15] Duan J, Tao C, Huang Y. Numerical analysis of cured-in-place pipe structural liner for underground pipeline rehabilitation. Comput Civ Eng 2023 2023:772–80.
[16] Rueda F, Marquez A, Otegui JL, Frontini PM. Buckling collapse of HDPE liners: experimental set-up and FEM simulations. ThinWalled Struct 2016;109:103–12.
[17] Liu Y, Nayak S. Structural health monitoring: state of the art and perspectives. JOM 2012;64:789–92. https://doi.org/10.1007/s11837-012-0370-9.
[18] Gharehbaghi VR, Noroozinejad Farsangi E, Noori M, Yang TY, Li S, Nguyen A, et al. A critical review on structural health monitoring: definitions, methods, and

perspectives. Arch Comput Methods Eng 2022;29:2209–35. https://doi.org/10.1007/s11831-021-09665-9.
[19] Zhang Z, Tao C, Yin H, Nie W, Duan J, Huang Y, et al. MXene materials: developments and sensor applications in modern infrastructure. J Mater Sci 2025;60:11744–77. https://doi.org/10.1007/s10853-025-11135-7.
[20] Wen C, Hu S, Wang J, Fang H, Xue X. Fiber optic sensing technology in underground pipeline health monitoring: a comprehensive review. Struct Health Monit 2024:14759217241280904. https://doi.org/10.1177/14759217241280904.
[21] Adebola TO, Moore ID, Hoult NA. Use of optical fibers to investigate the performance of pressure pipe liners spanning across a ring fracture. Tunn Undergr Space Technol 2022;119:104229.
[22] Treitz J, Lan H, Moore ID, Hoult NA. Testing and analysis of PVC liners under simulated grouting pressure. Eng Struct 2022;250:113496.
[23] Liu Y, Moore ID, Hoult NA. Field monitoring of a corrugated steel culvert using multiple sensing technologies. J Pipeline Syst Eng Pr 2020;11:04020030. https://doi.org/10.1061/(ASCE)PS.1949-1204.0000477.
[24] Yoo D-H, Jang B-S, Yim K-H. Nonlinear finite element analysis of failure modes and ultimate strength of flexible pipes. Mar Struct 2017;54:50–72.
[25] Yang K, Fang H, Zhang X, Li B, Hu Q. Investigation of mechanical properties of corroded concrete pipes after cured-in-place-pipe (CIPP) rehabilitation under multi-field coupling. Tunn Undergr Space Technol 2022;128:104656.
[26] Mahmoodian M, Shahrivar F, Setunge S, Mazaheri S. Development of digital twin for intelligent maintenance of civil infrastructure. Sustainability 2022;14:8664.
[27] Wang X, Yang X, Duan J, Xu L, Khan MI, Tao C, et al. A digital twin integrated smart-liner for visualization monitoring of oil and gas pipeline infrastructure. Appl Energy 2025;400:126558.
[28] Wang T, Feng K, Ling J, Liao M, Yang C, Neubeck R, et al. Pipeline condition monitoring towards digital twin system: a case study. J Manuf Syst 2024;73:256–74.
[29] Ritto TG, Rochinha FA. Digital twin, physics-based model, and machine learning applied to damage detection in structures. Mech Syst Signal Process 2021;155:107614.
[30] Hielscher T, Khalil S, Virgona N, Hadigheh SA. A neural network based digital twin model for the structural health monitoring of reinforced concrete bridges. Structures, 57. Elsevier; 2023, 105248.
[31] Wang S, Lin Y, Duan J, Yan H, Wang X, Xiong X, et al. Reinforcement response prediction of composite-concrete beams with crack patterns and deep learning. Comput Aided Civ Eng 2025:mice.70174. https://doi.org/10.1111/mice.70174.
[32] Karyofyllas G, Giagopoulos D, Jia X, Papadimitriou C. A digital twin-driven machine learning framework for structural condition monitoring using multi-datasets. Struct Health Monit 2025:14759217251324110. https://doi.org/10.1177/14759217251324110.
[33] Hoult NA, Ekim O, Regier R. Damage/deterioration detection for steel structures using distributed fiber optic strain sensors. J Eng Mech 2014;140:04014097. https://doi.org/10.1061/(ASCE)EM.1943-7889.0000812.
[34] Lan H, Ma G, Hoult NA, Moore ID. Stress concentrations due to simulated corrosion pits in buried metal pipes under longitudinal bending. J Civ Struct Health Monit 2022;12:785–96. https://doi.org/10.1007/s13349-021-00522-8.
[35] Gresnigt AM, Karamanos SA. Local buckling strength and deformation capacity of pipes (ISOPE). ISOPE International Ocean and Polar Engineering Conference. ISOPE-I; 2009.
[36] Fu G, Shannon B, Rathnayaka S, Deo R, Kodikara J. State of the art literature review on CIPP liners. Melbourne, Australia: Monash Univ; 2020.
[37] Stewart HE, O'rourke TD, Wham BP, Netravali AN, Argyrou C, Zeng X, et al. Performance testing of field-aged cured-in-place liners (CIPL) for cast iron piping. Ithaca, NY: Cornell Univ; 2015.
[38] Wang WC, Zhou B, Xu SH, Yang ZM, Zhang QY. Recent advances in soft optical glass fiber and fiber lasers. Prog Mater Sci 2019;101:90–171.
[39] Herbers M, Richter B, Marx S. Rayleigh-based crack monitoring with distributed fiber optic sensors: experimental study on the interaction of spatial resolution and sensor type. J Civ Struct Health Monit 2024. https://doi.org/10.1007/s13349-024-00896-5.
[40] Tan X, Fan L, Huang Y, Bao Y. Detection, visualization, quantification, and warning of pipe corrosion using distributed fiber optic sensors. Autom Constr 2021;132:103953.
[41] Nagulapally P, Shamsuddoha M, Rajan G, Mohan M, Prusty BG. Distributed fiber optic sensor-based strain monitoring of a riveted bridge joint under fatigue loading. IEEE Trans Instrum Meas 2021;70:1–10.
[42] Duan J, Tao C, Huang Y. Finite Element Analysis of Structural Lining Materials for Pipeline Rehabilitation. Construction Research Congress 2024. Des Moines, Iowa: American Society of Civil Engineers; 2024. p. 671–8. https://doi.org/10.1061/9780784485279.067.
[43] Ramberg W., Osgood W.R. 1943. Description of stress-strain curves by three parameters.
[44] Zhu X, Lv Z, Shuai J, Shi L, Yu S, Xia G. Ductile Fracture Criterion Parameter Calibration and Analysis: X80 Pipe Steel. Pressure Vessels and Piping Conference, 88506. American Society of Mechanical Engineers; 2024.
[45] Abdelmoety AK, Kainat M, Yoosef-Ghodsi N, Li Y, Adeeb S. Strain-based reliability analysis of dented pipelines using a response surface method. J Pipeline Sci Eng 2022;2:29–38.
[46] Wang X, Lin Z. Morphologic and synergistic effects of GNP/NS binary-filler-based multifunctional coatings with robust anti-corrosion and hydrophobic properties. Prog Org Coat 2021;157:106286.
[47] Fu G, Shannon B, Azoor R, Deo R, Kodikara J. Equations for gap-spanning design of underground cast iron pipes lined with thermosetting polymeric liners. Tunn Undergr Space Technol 2022;119:104234.
[48] Bisagni C. Numerical analysis and experimental correlation of composite shell buckling and post-buckling. Compos Part B Eng 2000;31:655–67.
[49] Graciano C, Casanova E, Martínez J. Imperfection sensitivity of plate girder webs subjected to patch loading. J Constr Steel Res 2011;67:1128–33.
[50] Liang P, Wang W, Yuan X, Liu S, Zhang L, Cheng Y. Intelligent fault diagnosis of rolling bearing based on wavelet transform and improved ResNet under noisy labels and environment. Eng Appl Artif Intell 2022;115:105269.
[51] Resnet18 — Torchvision main documentation n.d. ⟨https://docs.pytorch.org/vision/main/models/generated/torchvision.models.resnet18.html⟩ (accessed December 16, 2025).
[52] Tan X, Bao Y. Measuring crack width using a distributed fiber optic sensor based on optical frequency domain reflectometry. Measurement 2021;172:108945.
[53] Amidror I. Scattered data interpolation methods for electronic imaging systems: a survey. J Electron Imaging 2002;11:157–76.
[54] Zhang K, Chavez Torres JL, Zang Z. Numerical analysis of pipelines settlement induced by tunneling. Adv Civ Eng 2019;2019:4761904. https://doi.org/10.1155/2019/4761904.
[55] Zhao Y, Ma P, Chen Y, Liu K, Yan X, Ma B, et al. Stability assessment of CIPP liner under varied boundary conditions: a theoretical and simulation study. Results Eng 2024;22:102187.
[56] Wen C, Hu S, Wang J, Fang H, Xue X. Fiber optic sensing technology in underground pipeline health monitoring: a comprehensive review. Struct Health Monit 2024:14759217241280904. https://doi.org/10.1177/14759217241280904.
[57] Farrag K. 2015. Evaluation of structural liners for the rehabilitation of liquid and natural gas piping systems.
[58] Wang X, Duan J, Yang X, Yan H, Wang S, Tao C, et al. Dual-purpose smart liner system for oil and gas pipelines: digital twin-enabled rapid monitoring during strike events. Energy 2025:139397.